Memo 2: PRESSURE DROP AND REACTOR CONFIGURATION **Due Date: 06/04/2026**
Covering Chapter 4 – Part 2

OVERVIEW

Pressure drop considerations are significant in the design of fixed-bed reactors. Exit pressures are often constrained by required conditions in downstream separations operations and increases in inlet pressure often come at a cost (considering the cost of compression or pumping). A design guideline used for pressure drop in the specification of fixed-bed reactors is that the pressure drop should never exceed 1/10 of the initial pressure ($\Delta P \leq 0.1\ P0$). Because pressure drop tends to increase in a bed over the life of the catalyst due to particle attrition and breakage, this guideline should be used as an absolute upper limit to design for pressure drop. Obviously, the lower the pressure drop the better.

The reactor of choice for these reactions we are focusing on in this project is the multi-tubular fixed bed reactor. For highly exothermic reactions, multi-tubular reactors offer the advantage of good heat transfer due to the small diameter of individual tubes, and the opportunity to optimize pressure drop by manipulating the number of tubes needed to achieve your target reactor volume. Remember that this reactor will need to be fabricated, transported and installed, so practical reactor length should be considered.

For Memo 2 you developed a simple model for an isothermal reactor using a single reaction and a single mole balance equation using POLYMATH and /or Python, and you developed a simple plug flow model using Aspen Plus. In this memo, you will develop a Python model to include mole balance and momentum balance (pressure drop equation) using the Ergun equation (see Fogler) and then explore the relationship between pressure drop and number of tubes. You can also expand your POLYMATH model to do the same (OPTIONAL).

PART 1: SINGLE-PIPE REACTOR OPTIMIZATION (Python)

Present a preliminary design of your reactor based on the kinetics used in Memo 2. Use **Python** for this portion of the reactor design. Your model **must** contain at least **one mole balance** and a **momentum balance** (pressure drop) using the Ergun equation.

Start with a single pipe reactor. Assume that you have an isothermal reactor with the approximate reactor volume determined in Memo 2. Create the plots of conversion vs. volume (or catalyst weight) for contours of constant temperatures for your selected temperature range, as done in Memo 2. Select a catalyst particle size and bed void fraction based on your literature search and document this reference. (If you don't have values for these, 5mm particles with 0.4 void fraction is a good place to start).
Comment on the differences when pressure drop is a consideration.

Hint-1: Double check your units. Many students make a simple mistake in the units resulting in either very small pressure drop, or very high pressure drop (polymath will not work)

Hint-2: For your first guess of your reactor's diameter and length, start with what we know: $3 < L/D < 5$, and diameter should not exceed 8m.

- Start with $L/D = 5$. If $y > 0.9$, then your design should be fine. Keep an eye on the value of D. If in this case ($L/D = 5$) D is > 8m, you can increase L/D to maintain D to be < 8m.
- If $y < 0.9$ (high pressure drop), decrease L/D. Keep decreasing L/D until $y > 0.9$ (even if you have to go for $L/D < 3$)

PART 2: MULTI-TUBE CONFIGURATION ANALYSIS

Once you know the reactor length at which y > 0.9, now you can split the single pipe into multiple tubes. (Don't forget that you will need to divide the total volume into multiple tubes. Also, the flow-rates of all species will have to be divided by number of tubes). Knowing the total volume and tube length, explore the relationship between the number of tubes and pressure drop.

$$V = n_t \frac{\pi}{4} D_t L$$

- Assume 1" tubes, explore the relationship between pressure drop and tube length for different tubes number (total volume is constant). You should see that if you use too few tubes (very long tubes) pressure drop will increase to a point that polymath won't work. ***Discuss your results.***
- Keeping the value of L (obtained in part 1) constant, and for a constant total volume, explore the relationship between pressure drop and tubes sizes for different tubes number. ***Discuss your results.***
- Keeping the value of L (obtained in part 1) constant, and for a constant total volume, and assuming 1" tubes, Plot ΔP vs. L. For comparison, increase your particle size by a factor of two and create the same ΔP vs. L plot on the same graph. ***Discuss your results.***

Include references for all data (conversion used, inlet T and P, approximate reactor volume etc.).

PART 3: SENSITIVITY ANALYSIS

- How does varying inlet pressure affect pressure drop?
- How sensitive is pressure drop to catalyst particle size?
- How much does bed porosity affect pressure drop?
- How sensitive is your design to L/D ratio choice? (Part 1)

PART 4: POLYMATH VERIFICATION (OPTIONAL but Highly recommended)

If interested in verification:

1. Set up equivalent ODE model in POLYMATH
2. Use same kinetics and parameters as Python
3. Compare W_design and ΔP at design point
4. Results should match within 1%

PART 5: ASPEN PLUS SIMULATION (OPTIONAL)

If interested in industry software:

1. Create RPLUG model with Ergun pressure drop
2. Run single-tube configuration from Part 1
3. Compare with Python and POLYMATH
4. Include model description and results

Late submission: Less than 24 hrs: 50% credit only
More than 24hrs: 0% credit